SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Aravind M. (192126006 8) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Aravind M.

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	1.4	1.4	1.4
Concept Mapping Structure	25%	1.75	1.75	1.75
Linking & Relationships	20%	1.4	1.4	1.4
Application & Analysis	20%	1.4	1.4	1.4
Explanation & Clarity	15%	1.05	1.05	1.05

Total Marks 21 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2 Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:
DataLink→MACAddress→LocalDelivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:
ErrorControl
→DataLinkLayer→ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:
Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗
Analyze the problem and explain how the concept map helps in troubleshooting.

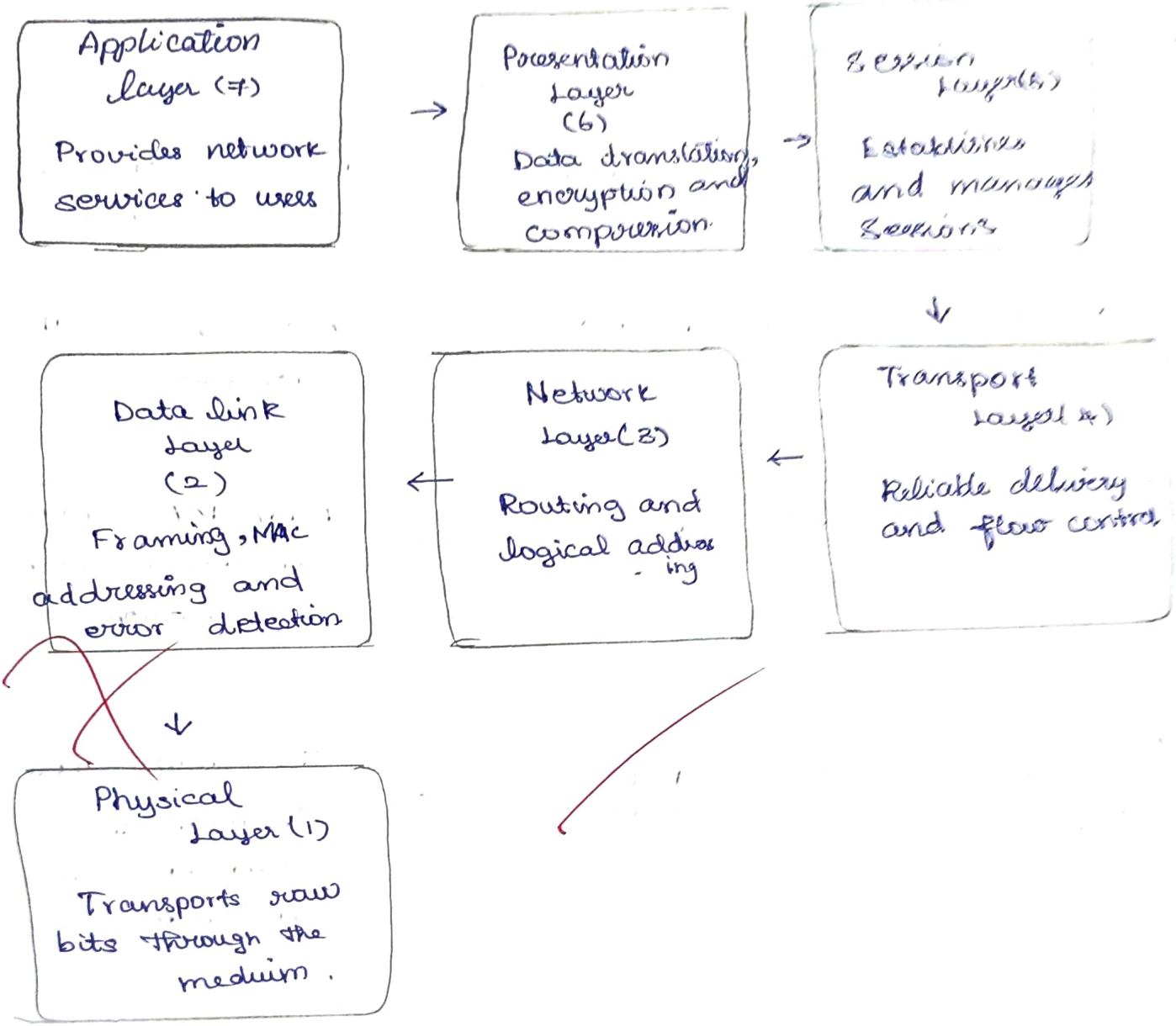
RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Computer networks.

Q1. Identify the missing OSI layers and explaining their functions.



Missing layers:

- session layer (Layer 5): Establishes, manages and terminates communication session between devices.
- Data link layer (layer - 2) - Transfers data transfers between directly connected devices and provides error detection.

Q2. Construct a concept map linking data units with their respective OSI layers and analyze how transformation supports transmission.

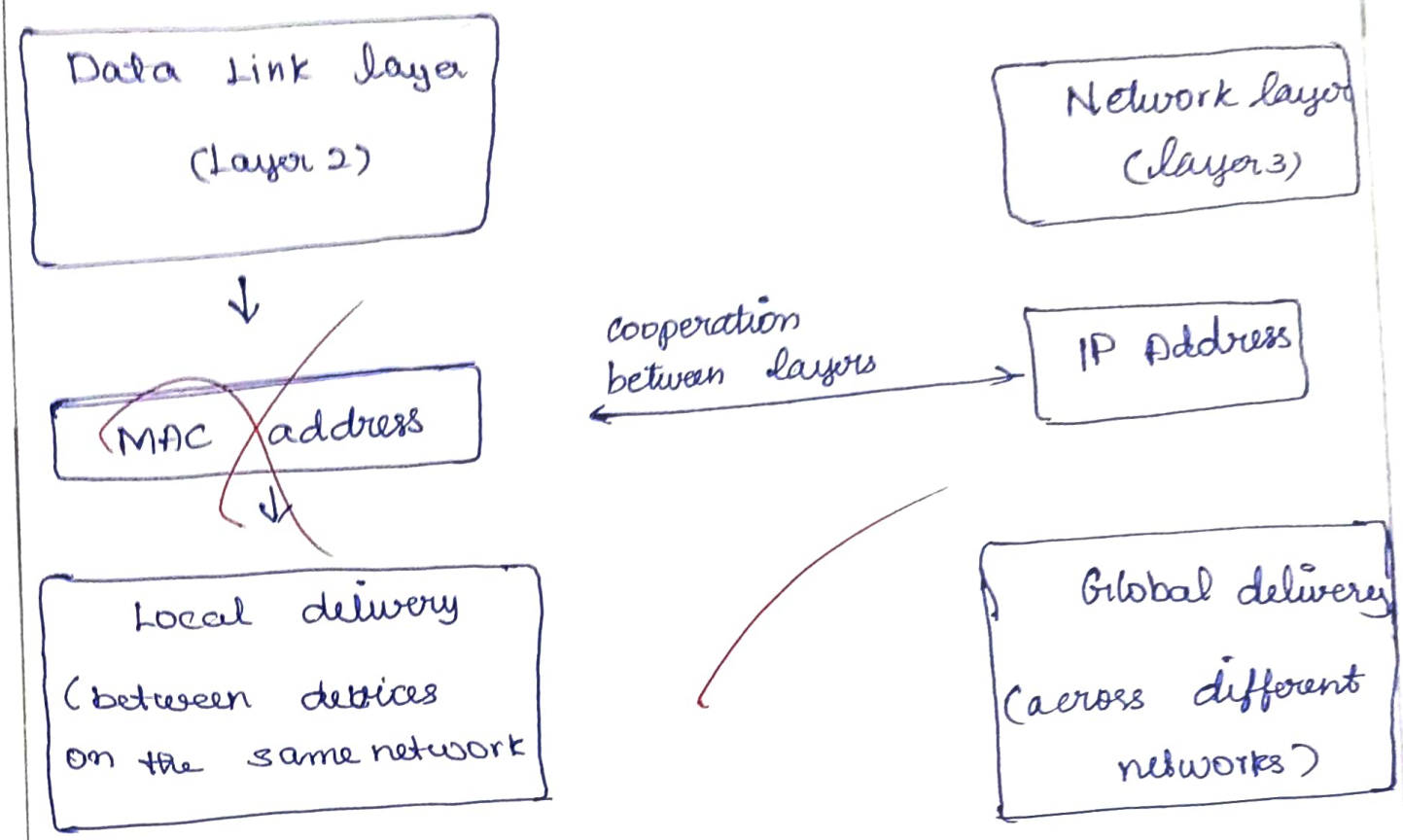
Data Units and OSI layers:

Transport layer (layer 4)	→	Segment	→ Divides data into manageable units
Network layer (Layer 3)	→	Packet	→ Adds logical IP addressing.
Data link layer (Layer 2)	→	Frame	→ Adds MAC addressing and error checking information.
Physical layer (layer-1)	→	Bits (0s and 1s)	→ Transmits data through the communication medium.

Transformation:

- Encapsulation (Sender side) - data is converted into different data Units as it moves down
- Decapsulation (reciever side) - data is converted back to its original form.

Q8. Explain how data link → MAC address → Local delivery and Network → IP Address → Global delivery ensure complete data delivery.



Example:

When a student opens a actions. the Network layer ~~thighs~~ helps packets towards the web sever using IP addresses.